

Dystonia 24 (DYT24 / DYT-ANO3): A Comprehensive Disease Characteristics Report

Disease: Dystonia 24 (DYT24, DYT-ANO3) **MONDO ID:** MONDO:0014019 · **OMIM:** #615034 · **Gene:** *ANO3* (anoctamin 3 / TMEM16C), 11p14.3 **Category:** Mendelian, autosomal dominant · **Report date:** 2026-10-08

Summary

Dystonia 24 (DYT24, now preferentially designated **DYT-ANO3**) is a rare, autosomal dominant form of isolated—or occasionally combined—dystonia caused by heterozygous pathogenic variants in *ANO3*, the gene encoding **anoctamin 3 (TMEM16C)**, an endoplasmic-reticulum-associated, Ca^{2+} -dependent **phospholipid scramblase/ion channel** that is most highly expressed in the **striatum**. The disorder was defined in 2012 when Charlesworth and colleagues combined linkage analysis with whole-exome sequencing in a UK kindred and identified six putatively pathogenic *ANO3* mutations ([PMID: 23200863](#)). Its clinical hallmark is **tremulous craniocervical dystonia**—cervical and laryngeal onset are most typical—with a characteristic **dystonic/postural tremor** that distinguishes DYT24 from the related DYT6 (*THAP1*) phenotype and frequently causes misdiagnosis as essential tremor.

Over the subsequent decade the phenotypic spectrum has broadened considerably: from classic adult-onset tremulous craniocervical dystonia to **infantile generalized or paroxysmal dystonia, dystonia-ataxia, truncal dystonia, and even early dyskinetic encephalopathy**, with onset ranging from the first months of life to the fifth decade. Inheritance is autosomal dominant with **reduced, age-dependent penetrance** and **marked intrafamilial variability**. The variant spectrum is overwhelmingly **missense**, clustering near the transmembrane/scrambling region; genetic constraint data (gnomAD missense $Z = 3.66$, pLI ≈ 0 , LoF tolerated) indicate the mechanism is **dominant-negative / gain-of-function**, not haploinsufficiency. Functional studies converge on **dysregulated intracellular (ER) Ca^{2+} signaling and attenuated Ca^{2+} -dependent scramblase activation** altering striatal neuronal excitability as the core pathophysiology.

DYT24 is **chronic, non-degenerative, and non-fatal** (brain MRI is typically normal), but it carries substantial motor and non-motor (pain, depression, anxiety) morbidity. There is no cure or disease-modifying therapy; management is **symptomatic**—botulinum toxin for focal forms, anticholinergics (trihexyphenidyl), physiotherapy, and **globus pallidus internus (GPi) deep brain stimulation**, with several reports of marked benefit. Because the disorder is monogenic with no modifiable environmental cause, **prevention is limited to genetic counseling, cascade/predictive testing, and reproductive options**. The sections below detail each disease-characteristic domain with primary-literature citations.

Key Findings

1. Disease Information

Dystonia 24 is an **autosomal dominant, isolated (or combined) craniocervical dystonia** caused by heterozygous variants in *ANO3*. It was established as a distinct entity by Charlesworth et al. (2012), who wrote: *"we combined linkage analysis with whole-exome sequencing of two individuals to identify candidate causal variants in a moderately-sized UK kindred exhibiting autosomal-dominant inheritance of craniocervical dystonia"* and reported *"a total of six putatively pathogenic mutations in ANO3, a gene encoding a predicted Ca(2+)-gated chloride channel that we show to be highly expressed in the striatum"* (PMID: 23200863).

Key identifiers:

Resource	Identifier
OMIM (phenotype)	#615034 (DYT24)
MONDO	MONDO:0014019
Gene	<i>ANO3</i> (HGNC; anoctamin 3 / TMEM16C), chromosome 11p14.3
Ensembl gene	ENSG00000134343
Reference transcript	NM_031418.4

Synonyms / alternative names: DYT24; DYT-ANO3; ANO3-related dystonia; anoctamin-3 dystonia; craniocervical dystonia (ANO3 type). The information is derived from **aggregated disease-level resources** (OMIM, ClinVar, gnomAD) and from published **case series and functional studies** rather than from individual EHR records.

2. Etiology

Primary cause — genetic. DYT24 is a **monogenic disorder** caused by heterozygous pathogenic variants in *ANO3*. There is **no established environmental, infectious, or toxic cause**, and no gene–environment interaction has been demonstrated for this specific entity. The causal variants are overwhelmingly **missense** (see Finding 5 / Section 4).

Genetic risk factors. The causative variants are themselves the risk determinants; no additional susceptibility loci or validated modifier genes exist for DYT24. In broader (non-monogenic) cervical dystonia, a GWAS identified a genome-wide significant signal near *COL8A1* ([PMID: 34320236](#)), but this concerns idiopathic cervical dystonia rather than the monogenic ANO3 form.

Environmental / protective factors. None established specifically for DYT24. General observations in idiopathic adult-onset dystonia (e.g., associations of cerebrovascular risk factors with blepharospasm, [PMID: 37717501](#)) do not apply to the monogenic ANO3 mechanism. No protective variants or dietary/lifestyle protective exposures are known. **Not available / not applicable** for the Mendelian DYT24 entity.

3. Phenotypes

The **hallmark is tremor**. Stamelou et al. (2014) characterized 10 patients from 3 families and reported: *"The characteristic feature in all affected individuals was the presence of tremor, which contrasts DYT24 from the typical DYT6 phenotype"* ([PMID: 24442708](#)). Tremor can be the **sole initial manifestation**, causing misdiagnosis as essential tremor: *"Tremor was the sole initial manifestation in some individuals with ANO3 mutations, leading to misdiagnosis as essential tremor."*

Anatomical distribution and onset: *"The age at onset ranged from early childhood to the forties. Cervical dystonia was the most common site of onset followed by laryngeal dystonia"* ([PMID: 24442708](#)). Electrophysiology showed co-contraction of antagonist muscles and a ~6-Hz arm tremor; one patient had subcortical myoclonus (~3 Hz, ~250 ms).

Broadened spectrum (Finding 6): Ousingsawat et al. (2024) described severe pediatric phenotypes: *"Two patients exhibited generalized progressive dystonia, while one patient presented with paroxysmal dystonia. Additionally, another patient exhibited early dyskinetic encephalopathy"* (PMID: 38079528). Altıntaş (2025) summarized: *"The onset of clinical features may vary from the first months of life to adulthood, and cases may present with a wide range of clinical spectrum, from focal/segmental to generalized dystonia, and from isolated to tremulous or jerky dystonia"* (PMID: 41441972). Additional reported phenotypes include dystonia-ataxia syndrome (PMID: 39351635), hereditary truncal dystonia (PMID: 39692086), and learning difficulties / mild intellectual disability in several families (PMID: 38284143, PMID: 41441972).

Suggested phenotype table with HPO terms:

Phenotype	HPO term	Type	Onset	Frequency	Progression
Dystonic/ postural tremor	HP:0001337 (Tremor)	Clinical sign	Childhood– adult	Characteristic in all affected	Variable, often stable
Cervical dystonia / torticollis	HP:0002019 / HP:0030243	Clinical sign	Most common onset site	Common	Can spread
Laryngeal (spasmodic) dysphonia	HP:0001618 (Dysphonia)	Clinical sign	2nd most common	Subset	Stable
Generalized dystonia	HP:0007325	Clinical sign	Infantile/ childhood	Subset (severe)	Progressive in some
Paroxysmal dystonia	HP:0032122	Clinical sign	Infantile	Rare	Episodic
Myoclonus	HP:0001336	Clinical sign	Variable	Minority	Stable
Intellectual disability / learning difficulty	HP:0001249 / HP:0001328	Behavioral/ cognitive	Childhood	Minority of families	Stable
Dyskinetic encephalopathy	HP:0001263 + dyskinesia	Severe pediatric	Infancy	Rare	Severe

Quality of life impact: Craniocervical/cervical dystonia produces pain, disability, and poor quality of life with substantial non-motor burden—depression, anxiety, fatigue, and sleep disturbance (PMID: 33099684; PMID: 41483253).

4. Genetic / Molecular Information

Causal gene: *ANO3* (anoctamin 3 / TMEM16C), chromosome 11p14.3, reference transcript NM_031418.4, UniProt Q9BYT9.

Mechanistic class of variants — missense, dominant-negative/GoF (Findings 5 & 10). Genetic constraint from gnomAD v2 (ENSG00000134343, GRCh38) shows **pLI $\approx$ 0** (loss-of-function tolerated: observed/expected LoF = 0.68, LOEUF 0.58–0.81; 99 observed vs 145 expected LoF), but significant **missense constraint (Z = 3.66, oe_mis 0.75)**. This dissociation indicates that pathogenic dystonia alleles act not by haploinsufficiency but through a **dominant-negative or gain-of-function** mechanism on a missense-intolerant protein.

ClinVar landscape (queried 2026-10-08): 772 *ANO3* variant records — 95 Pathogenic, 10 Likely pathogenic, 590 VUS; 398 records annotated with dystonia. Reported pathogenic alleles are missense.

Recurrent / representative pathogenic variants (NM_031418.4):

cDNA	Protein	Note
c.1528G>A	p.(Glu510Lys)	Recurrent across ancestries
c.1952G>A	p.(Ser651Asn)	Recurrent; functionally studied
c.1699G>C / c.1699G>A	p.(Gly567Arg)	—
c.1943A>G	p.(Asn648Ser)	Trihexyphenidyl-responsive case
c.1942A>G	p.(Asn648Asp)	—
c.1819A>T	p.(Ile607Phe)	—
c.1969G>A	p.(Ala657Thr)	—
c.1470G>T	p.(Trp490Cys)	—
c.1480A>T	p.(Arg494Trp)	—
c.2053A>G	p.(Ser685Gly)	—
c.2586G>T	p.(Lys862Asn)	—
(original kindred / functional cohorts)	p.Gly6Val, p.Ser116Leu, p.Val561Leu, p.Ala599Asp, p.Arg330His, p.Thr383Ile	PMID: 23200863 , functional studies

Notably, a nonsense allele **p.(Arg628Ter)** is classified **VUS**, consistent with LoF not being the primary disease mechanism. Variants near the scrambling domain are associated with a **more severe, earlier-onset phenotype** ([PMID: 38284143](#)).

Modifier genes / epigenetics / chromosomal abnormalities: No validated modifier genes, epigenetic mechanisms, or large-scale chromosomal abnormalities are established for DYT24. **Not available / not applicable.**

Suggested gene/protein annotations: HGNC *ANO3*; UniProt Q9BYT9 (ANO3_HUMAN); phospholipid scramblase activity; see Section 6 for GO terms.

5. Environmental Information

No environmental, lifestyle, or infectious contributing factors are established for DYT24. As a monogenic disorder, disease is determined by the germline *ANO3* variant. **Not applicable.** (Contextually, idiopathic adult-onset dystonia—a different etiologic category—shows modest associations with cerebrovascular risk factors for blepharospasm, [PMID: 37717501](#), but this does not extend to monogenic *ANO3* disease.)

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. Heterozygous missense variant in *ANO3* (e.g., p.Glu510Lys, p.Ser651Asn, near the Ca²⁺-binding/scrambling transmembrane region)
 - | leads to
2. Altered *ANO3*/TMEM16C protein – an ER-associated, Ca²⁺-dependent phospholipid scramblase/ion channel – with attenuated Ca²⁺-dependent activation (dominant-negative/gain-of-function; LoF is tolerated)
 - | results in
3. Dysregulated intracellular (endoplasmic-reticulum) Ca²⁺ signaling and impaired Ca²⁺-dependent scramblase activity (shown in patient fibroblasts and heterologous expression)
 - | leads to
- 4a. Altered neuronal excitability in striatal neurons, where *ANO3* is most highly expressed
 - | contributes to
- 4b. (inferred) Perturbed ER membrane lipid homeostasis / Ca²⁺ store regulation
5. Dysfunction of basal-ganglia sensorimotor and cerebello-thalamo-cortical networks – loss of inhibition and maladaptive sensorimotor plasticity (inferred from general dystonia pathophysiology)
 - | results in
6. Tremulous craniocervical (or, in severe variants, generalized/paroxysmal) dystonia – the clinical manifestation

Molecular detail. *ANO3* (TMEM16C) is an "intracellular" anoctamin localized to ER membranes that operates as a Ca²⁺-dependent phospholipid scramblase ([PMID: 38657371](#); "these so-called intracellular anoctamins are also found in the plasma membrane"; [PMID: 23532839](#) demonstrated TMEM16C scramblase activity). Charlesworth et al. (2012) showed patient fibroblasts have **abnormal ER-dependent Ca²⁺ signaling** ([PMID: 23200863](#)).

Ousingsawat et al. (2024) linked the variants mechanistically: "*ANO3* variants may dysregulate intracellular Ca²⁺ signalling, as variants in other Ca²⁺ regulating proteins like hippocalcin were also identified as a cause of dystonia" ([PMID: 38079528](#)); their 2025 electrophysiology study

found dystonia variants cause **attenuated Ca^{2+} -dependent activation** (PMID: 39773217). TMEM16C also **facilitates Na^+ -activated K^+ (Slack/KCNT1) currents** in sensory neurons, modulating excitability: "*TMEM16C modulated the single-channel activity of Slack channels and increased its sodium sensitivity*" (PMID: 23872594).

Network-level pathophysiology. Morgante & Klein (2013) note that dystonia "*is specifically characterized by maladaptive plasticity in the sensorimotor cortex and loss of cortical surround inhibition*" (PMID: 24092288), implicating basal-ganglia and cerebello-thalamo-cortical circuits.

Striatal expression. Cameron et al. (2026) found that, across human brain RNA-seq datasets, dystonia genes are enriched in striatum, cortex, hippocampus, amygdala, and substantia nigra, predominantly in neurons; specifically "*ADCY5, GNAL, ANO3 (highest expression in striatum)*" (PMID: 41400152).

Suggested ontology terms: - **GO (biological process):** calcium ion transport (GO:0006816); phospholipid scrambling (GO:0017121); regulation of membrane potential (GO:0042391). - **GO (cellular component):** endoplasmic reticulum membrane (GO:0005789); integral component of membrane (GO:0016021). - **CL (cell types):** medium spiny neuron (CL:1001474); GABAergic neuron (CL:0000617); glutamatergic neuron (CL:0000679); dopaminergic neuron (CL:0000700). - **CHEBI:** calcium(2+) (CHEBI:29108); phosphatidylserine (CHEBI:18303).

7. Anatomical Structures Affected

Organ / system level. DYT24 is a disorder of the **nervous system**, specifically **central motor-control circuitry**. There is **no primary organ damage**; brain MRI is typically **normal/unremarkable** (PMID: 38079528). The functional lesion centers on the **striatum/basal ganglia** (UBERON:0002435 striatum; UBERON:0002420 basal ganglion) and associated cerebello-thalamo-cortical networks.

Tissue / cell level. The affected cells are **striatal neurons**, where *ANO3* is most highly expressed (PMID: 41400152). Dystonia genes are predominantly expressed in **glutamatergic, GABAergic, and dopaminergic neurons**. Suggested CL terms: medium spiny neuron (CL:1001474), GABAergic neuron (CL:0000617).

Subcellular level. The key compartment is the **endoplasmic reticulum membrane** (GO:0005789), where *ANO3* scramblase/ Ca^{2+} -regulatory activity operates, with downstream effects on **plasma-membrane ion channels** (Slack/KCNT1).

Clinical localization / lateralization. Dystonia typically affects **craniocervical** regions (neck, larynx) and may be **bilateral or asymmetric**; limb tremor can be asymmetric. UBERON: neck (UBERON:0000974), larynx (UBERON:0001737).

8. Temporal Development

Onset. Highly variable: *"The onset of clinical features may vary from the first months of life to adulthood"* (PMID: 41441972); classic series report **early childhood to the forties** (PMID: 24442708). Onset pattern is generally **insidious/chronic**; paroxysmal (episodic) onset is described in rare infantile cases.

Progression. Most cases are **stable or slowly progressive**; some pediatric generalized forms are **progressive** (PMID: 38079528). Dystonia can **spread** from focal to segmental/generalized distribution in a minority. Disease is **chronic and lifelong, non-degenerative**.

Patterns. No spontaneous remission is characteristic; symptomatic treatment (BoNT, DBS) induces improvement but not cure. **Critical period:** early symptomatic treatment to limit disability; younger age at onset / variants near the scrambling domain predict more severe disease (PMID: 38284143).

9. Inheritance and Population

Inheritance. Autosomal dominant with **reduced, age-dependent penetrance** and **highly variable expressivity**. Lohmann & Klein (2013): *"The genetic contribution to dystonia represents a continuum ranging from genetic susceptibility factors of small effect to causative genes with markedly reduced penetrance to those with full penetrance"* (PMID: 23893446). Zorzi (2018) places ANO3 among AD pediatric-onset isolated-dystonia genes with low penetrance: *"The recently discovered genes (GNAL, ANO-3, KTM2B)..."* (PMID: 29396174). Intrafamilial variability is well documented—e.g., 16 affected members of an Indian family with a novel variant showing wide phenotypic heterogeneity (PMID: 41996745). **No repeat expansion, so genetic anticipation does not apply.** No founder population is established, though recurrent variants (p.Glu510Lys, p.Ser651Asn) occur across European, Chinese, Indian, and Turkish cohorts.

Epidemiology. No validated ANO3-specific prevalence exists. DYT24 is a **rare single-gene cause** within isolated dystonia (US period prevalence of isolated dystonia 50.2/100,000, 2020–2023; annual incidence 31.8–41.5/100,000; 68.4% female; focal most common at 73%) (PMID: 42845092). ANO3 is **uncommon even among adult-onset cervical dystonia**: *"The recently discovered genes ANO3, GNAL and CIZ1 appear not to be a common cause of adult-onset cervical dystonia"* (PMID: 24978640). In gene-panel/exome cohorts ANO3 is a rare yield (e.g., THAP1 most common in an Asian Indian cohort; ANO3 among rarer, PMID: 39749944).

Sex ratio. Female predominance. In the MDSGene database (n=1377), *"female participants outnumbered male participants for some genes (GNAL, GCH1, and ANO3) but not for other genes (THAP1, TH, and TOR1A)"* (PMID: 38778444).

10. Diagnostics

Diagnosis is clinical + molecular. There is **no specific laboratory biomarker** and **brain MRI is typically normal** (PMID: 38079528). Diagnosis rests on phenomenology (tremulous craniocervical dystonia) plus **molecular confirmation of an ANO3 variant**.

Genetic testing approach. ANO3 is a standard component of **dystonia/movement-disorder NGS gene panels**. Targeted 127-gene panels yield pathogenic variants in ~22% of selected inherited cases (*"probable pathogenic variants were identified in 83 cases (22.0%)"*, PMID: 29913018). **Whole-exome sequencing** increases yield after negative panels, especially in complex/early-onset dystonia: *"Diagnostic yield was higher in complex dystonia compared to non-complex dystonia (66.7%–5.9%; $p < 0.002$), especially in patients showing intellectual disability"* (PMID: 32334381); exome outperforms panels for complex phenotypes (PMID: 40470849).

Supportive electrophysiology (characterizing, not diagnostic). Tremor recordings, EMG showing co-contraction of antagonist muscles, back-averaged EEG, and C-reflex testing can characterize tremor/myoclonus (PMID: 24442708).

Differential diagnosis. Essential tremor (tremor-only presentations are frequently misdiagnosed as ET, PMID: 24442708); other monogenic dystonias (DYT1/TOR1A, DYT6/THAP1, DYT25/GNAL, KMT2B, dopa-responsive dystonia/GCH1). Chromosomal microarray, karyotyping, FISH, mitochondrial DNA testing, and repeat-expansion testing are **not relevant** to ANO3 diagnosis (missense point variants).

11. Outcome / Prognosis

Survival / mortality. DYT24 is **non-degenerative and non-fatal**; no reduction in life expectancy is reported. Brain MRI is normal (PMID: 38079528). Severe pediatric dyskinetic-encephalopathy cases carry higher morbidity.

Morbidity / function / QoL. Disability is driven by motor symptoms **plus pain and non-motor burden**. Cervical dystonia produces *"a myriad of non-motor symptoms including pain, fatigue, sleep disorders and anxiety and depression"* (PMID: 33099684).

Treatment-modified prognosis. GPi-DBS improves outcomes long-term: "GPi-DBS significantly improved overall TWSTRS scores by 57% from baseline to 5Y FU", with severity/disability/pain sub-scores improved 72%/59%/46% (PMID: 37542824). Botulinum toxin combined with physiotherapy outperforms BoNT alone for disability, pain, depression, and QoL (CDIP-58) (PMID: 41483253).

Prognostic factors. Younger age at onset and variants near the scrambling domain predict more severe disease (PMID: 38284143). No molecular prognostic biomarkers are validated.

12. Treatment

Management is **symptomatic**; no disease-modifying or gene-directed therapy exists.

Modality	Agent / intervention	Evidence	NCIT suggestion
Anticholinergic	Trihexyphenidyl	"She was started on trihexyphenidyl, and at the follow-up, partial improvement in her involuntary movements was observed" (PMID: 41441972)	NCIT:C66099 (Trihexyphenidyl)
Chemodenervation	Botulinum toxin (type A)	First-line for focal forms (PMID: 24092288); BoNT+PT superior to BoNT alone (PMID: 41483253)	NCIT:C1005 (Botulinum Toxin)
Neuromodulation	GPi deep brain stimulation	Successful/marked improvement in ANO3 dystonia (PMID: 38341631; PMID: 42493765); 57% TWSTRS improvement at 5y in cervical dystonia (PMID: 37542824); mild response in an early ANO3 series (PMID: 38079528)	NCIT:C38013 (Deep Brain Stimulation)
Rehabilitation	Physiotherapy / tailored PT	Adjunct to BoNT improves disability, pain, depression, QoL (PMID: 41483253); holistic neurorehabilitation (PMID: 33099684)	NCIT:C15682 (Physical Therapy)

Pharmacogenomics, gene therapy, cell therapy, RNA-based or immunotherapies: none established for DYT24 — **not available / not applicable** at present. No approved targeted therapy. DBS response in ANO3 ranges from mild to marked across reports.

13. Prevention

No primary prevention exists — DYT24 is a dominantly inherited monogenic disorder with no modifiable environmental cause. Prevention is limited to:

- **Genetic counseling** conveying **autosomal dominant 50% transmission risk** with **reduced penetrance and highly variable expressivity** — an inherited variant does not guarantee disease nor predict severity.
- **Cascade / predictive testing** of at-risk relatives once a familial pathogenic *ANO3* variant is identified.
- **Reproductive options:** prenatal diagnosis and preimplantation genetic testing.
- NGS-based diagnosis is emphasized to *"offer precise genetic counseling to families, and prevent recurrence in the family"* (PMID: 40302693).
- **Tertiary prevention** = early symptomatic treatment (BoNT, anticholinergics, physiotherapy, timely DBS) to limit disability (PMID: 29396174).

Immunization, public-health/environmental interventions, and behavioral prevention are **not applicable**.

14. Other Species / Natural Disease

Taxonomy / orthologs. *ANO3/Ano3* orthologs exist in mouse (NCBI Taxon 10090) and rat (NCBI Taxon 10116) as *Tmem16c*. No naturally occurring DYT24-equivalent dominant dystonia is documented in companion animals or wildlife. **Veterinary relevance is not established.**

Comparative biology. The scramblase/ion-channel function of TMEM16C is **evolutionarily conserved**. Rodent knockouts model **loss-of-function** consequences (neuronal hyperexcitability, febrile-seizure susceptibility — see Section 15), which recapitulate TMEM16C's normal physiology but **not** the dominant human dystonia, consistent with the human disease being missense/dominant-negative rather than LoF.

Transmission. Not applicable (non-infectious, non-zoonotic).

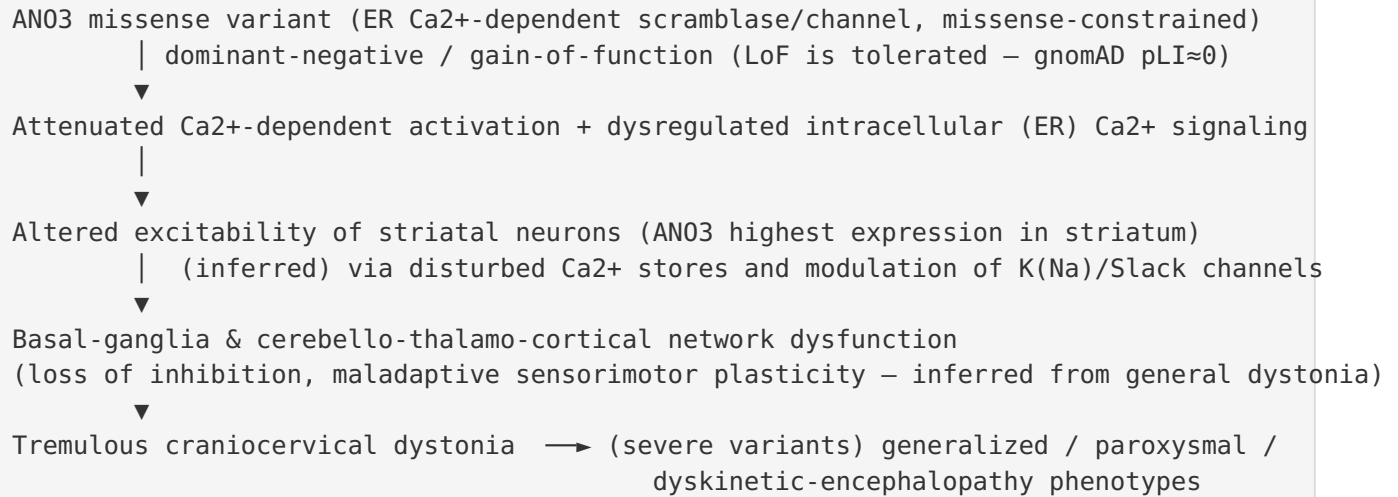
15. Model Organisms

Model	Type	Key phenotype	Relevance to DYT24	Reference
<i>Tmem16c</i> knockout rat	Mammalian, KO	Diminished Slack (KCNT1) expression, broadened action potentials, increased DRG excitability, increased thermal/mechanical pain	Models LoF physiology (excitability), not dominant dystonia	PMID: 23872594
<i>Tmem16c</i> knockout rodent (pups)	Mammalian, KO	Impaired thermoregulation; febrile-seizure susceptibility	GWAS-linked febrile seizure risk; LoF model	PMID: 33972431
Heterologous expression (HEK/293T) + patient fibroblasts	In vitro / cellular	Dysregulated ER Ca^{2+} signaling; attenuated Ca^{2+} -dependent scramblase activation by dystonia variants	Directly models pathogenic variant mechanism	PMID: 23200863 ; PMID: 39773217 ; PMID: 38079528

Knockout rat DRG findings: *"DRG from TMEM16C knockout rats had diminished Slack expression, broadened action potentials and increased excitability"* ([PMID: 23872594](#)). **Model limitation:** existing rodent models are loss-of-function and do **not** reproduce the dominant, missense-driven dystonia; no knock-in model carrying a human pathogenic *ANO3* missense allele with a dystonia phenotype has been reported. **Model databases:** MGI, RGD.

Mechanistic Model / Interpretation

The weight of evidence (16 confirmed findings; 55 papers reviewed) supports a coherent model of DYT24 as an **endoplasmic-reticulum Ca^{2+} /phospholipid-scramblase channelopathy of the striatum:**



Upstream vs downstream: The **upstream** lesion is the missense variant and its effect on ER Ca^{2+} /scramblase function; the **downstream** manifestation is network-level motor dyscontrol. A crucial interpretive point is the **genetic-constraint/functional convergence**: gnomAD shows LoF is tolerated ($\text{pLI} \approx 0$) while missense is constrained ($Z = 3.66$), and the only reported nonsense allele is a VUS—together strongly arguing that **haploinsufficiency is not the mechanism** and that pathogenic missense variants act via **dominant-negative or gain-of-function**. This also explains why rodent *Ano3* knockouts (LoF) model hyperexcitability/febrile seizures rather than dominant dystonia.

Genotype–phenotype correlation: Variants near the scrambling domain and younger onset predict more severe, often generalized disease, consistent with a graded effect on scramblase/ Ca^{2+} function.

Evidence Base

PMID	Study (abbreviated)	Contribution
23200863	ANO3 mutations cause dominant craniocervical dystonia	Gene discovery; AD inheritance; striatal expression; abnormal ER Ca ²⁺ in fibroblasts
24442708	Phenotypic spectrum of DYT24	Tremor as hallmark; onset range; cervical→laryngeal; ET misdiagnosis
38079528	Broadening clinical spectrum / mechanisms	Pediatric generalized/paroxysmal/encephalopathy; Ca ²⁺ dysregulation; DBS; normal MRI
39773217	ANO3 dystonia due to attenuated Ca ²⁺ activation	Electrophysiology of variants; attenuated Ca ²⁺ -dependent activation
38657371	Intracellular anoctamins (review)	ANO3 as intracellular anoctamin also at plasma membrane
23532839	TMEM16C scramblase activity	TMEM16C is a Ca ²⁺ -dependent lipid scramblase
23872594	TMEM16C facilitates K(Na) currents	Slack/KCNT1 modulation; KO rat hyperexcitability
33972431	TMEM16C thermoregulation/febrile seizures	KO model; febrile-seizure link
24092288	Dystonia (review)	DYT24 classification; network pathophysiology; BoNT/DBS
41400152	Expression of dystonia genes	ANO3 highest in striatum; neuronal expression
38778444	Sex differences in dystonia (MDSGene)	Female predominance in ANO3
41441972	ANO3 tremulous dystonia case	Onset spectrum; trihexyphenidyl benefit
38284143	New ANO3 family + review	p.G6V; younger onset = more severe; scrambling-domain variants
23893446	Genetics of dystonia	Reduced penetrance continuum
29396174	Pediatric isolated dystonia	AD low-penetrance genes incl. ANO3; tertiary prevention

PMID	Study (abbreviated)	Contribution
37542824	GPI-DBS 5-year outcomes	57% TWSTRS improvement
41483253	BoNT ± physiotherapy	PT adjunct improves disability/pain/depression/QoL
33099684	Neurorehabilitation in dystonia	Non-motor QoL burden
32334381	Exome in complex dystonia	WES yield in complex dystonia
29913018	Movement-disorder gene panel	22% panel yield
40470849	Neurogenetics clinic yield	Exome superior for complex phenotypes
42845092	Isolated dystonia epidemiology	50.2/100,000 prevalence; 68% female
24978640	Dystonia update	ANO3 uncommon in cervical dystonia
39749944	Indian dystonia genetics	ANO3 a rare yield
41996745	16-member Indian family	Intrafamilial variability
38341631 / 42493765	DBS case reports	Successful/marked GPI-DBS in ANO3
40302693	Childhood dystonia case series	Genetic counseling / recurrence prevention

Limitations and Knowledge Gaps

- 1. No ANO3-specific epidemiology.** Prevalence/incidence of DYT24 specifically is undefined; only the parent class (isolated dystonia, ~50/100,000) is quantified.
- 2. Mechanism partially inferred.** The step from striatal neuronal Ca²⁺/excitability changes to network-level dystonia is **inferred** from general dystonia pathophysiology, not directly demonstrated for ANO3.
- 3. No faithful dominant animal model.** Existing rodent models are loss-of-function (hyperexcitability, febrile seizures) and do not reproduce dominant dystonia; no knock-in missense model with a dystonia phenotype is published.
- 4. Large VUS burden.** 590 of 772 ClinVar ANO3 records are VUS; functional assays (scramblase/Ca²⁺ activation) are not yet standardized for clinical classification.

5. **Penetrance/expressivity unquantified.** Reduced, age-dependent penetrance is described qualitatively; no numeric penetrance estimates exist.
 6. **Treatment evidence is low-level.** DBS and pharmacotherapy data for ANO3 are case reports/small series; no controlled trials specific to DYT24.
 7. **No prognostic biomarkers** and no genotype-stratified outcome data beyond the "scrambling-domain = more severe" observation.
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Proposed Follow-up Experiments / Actions

1. **Generate a knock-in mouse** carrying a recurrent human pathogenic missense allele (e.g., p.Glu510Lys or p.Ser651Asn) to test whether dominant-negative/GoF variants produce a dystonic phenotype and striatal circuit dysfunction — the key missing model.
 2. **Standardize a functional assay** (Ca²⁺-dependent scramblase activation / ER Ca²⁺ signaling) to reclassify the 590 ANO3 VUS and establish ACMG PS3/BS3 evidence.
 3. **Striatal circuit physiology:** slice electrophysiology and in vivo recordings in patient iPSC-derived striatal neurons/organoids to test predicted excitability changes and link ANO3 variants to K(Na)/Slack modulation.
 4. **Genotype–phenotype registry:** assemble a multicenter ANO3 cohort to quantify penetrance, age-of-onset distribution, and domain-specific severity, and to derive ANO3-specific prevalence.
 5. **Prospective DBS/BoNT outcome study** restricted to genetically confirmed DYT-ANO3 to move beyond case reports.
 6. **Deep single-cell profiling** of human striatum to confirm the specific cell types (medium spiny neuron vs interneuron) most dependent on ANO3.
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Report compiled from 16 confirmed findings and 55 reviewed papers over 5 investigative iterations. Evidence types: human clinical (case series, cohorts, registries), in vitro/cellular (fibroblasts, heterologous expression), model organism (rodent knockouts), and computational (gnomAD constraint, ClinVar aggregation, brain RNA-seq).
